ELEC 221 Lecture 12
The CT Fourier transform properties: convolution and multiplication

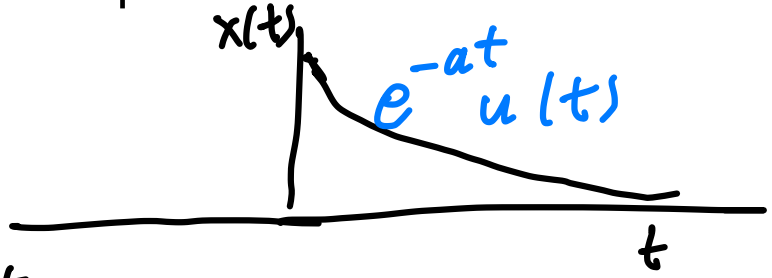
Tuesday 20 October 2026

Announcements

- Quiz 6 today
- Tutorial Assignment 3 due Friday 23:59

Exercise

If you finish quiz 6 early: compute the Fourier transform of $x(t) = e^{-at}u(t)$ $(a > 0)$.



$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} e^{-at}\, u(t)\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-at}\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-(j\omega + a)t}\, dt$$

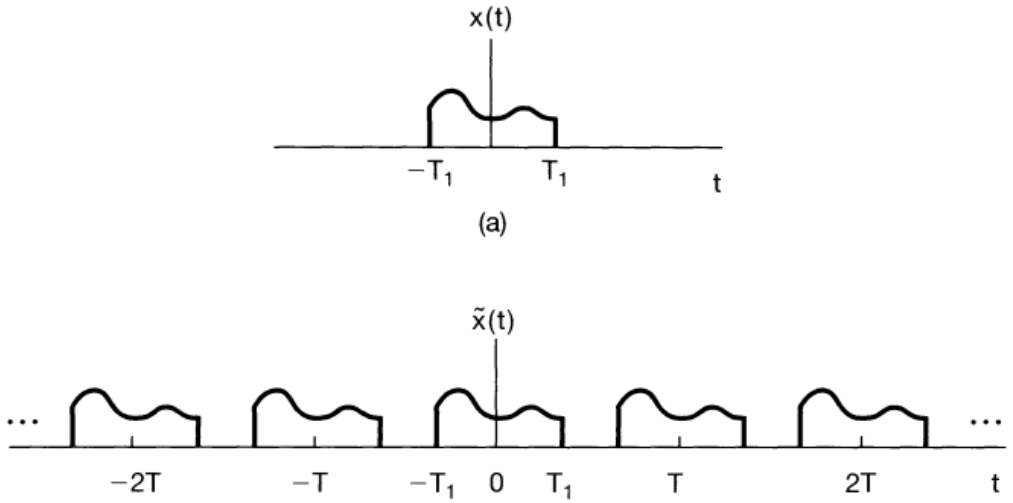
$$= -\frac{1}{j\omega + a}\, e^{-(j\omega + a)t}\Big|_{0}^{\infty}$$

$$= -\frac{1}{j\omega + a}\left[0 - 1\right]$$

$$= \frac{1}{a + j\omega}$$

Last time

We generalized the CT Fourier series (for periodic signals) to the Fourier transform (for aperiodic signals):



We expressed the periodic extension of an aperiodic function as

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

Image credit: Oppenheim chapter 4.1

Fundamental period N

c_{-2} c_{-1} c_0 c_1 c_2 c_3 c_4 -- c_{N-1} c_N c_{N+1}

same same

if $x[n]$ real,

$c_{-1} = c_1^*$ $\Rightarrow$

real signal $N = 6$

c_0 c_1 c_2 c_3 c_4 c_5

$c_4 = c_2^*$

$c_5 = c_1^*$

Last time

We computed its Fourier coefficients:

$$c_k = \frac{1}{T} \underbrace{\int_{-\infty}^{\infty} x(t) e^{-jk\omega t} dt}_{X(jk\omega)}$$

We put this into the Fourier series and let $T \to \infty$ ($\omega \to 0$):

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(jk\omega) e^{jk\omega t} = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega) e^{jk\omega t} \omega$$

$$\lim_{T \to \infty} \tilde{x}(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega = x(t)$$

Last time

Inverse Fourier transform (synthesis equation):

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(jw) e^{jwt} dw$$

Fourier transform (analysis equation):

$$X(jw) = \int_{-\infty}^{\infty} x(t) e^{-jwt} dt$$

Today

We will make a big step towards answering the question
"Why are we even doing this?"

Learning outcomes:

- Compute the Fourier transform of periodic functions
- Describe the time/frequency duality of the Fourier transform
- Apply the *convolution property* of the Fourier transform to characterize LTI system behaviour
- Describe the *multiplication property* of the Fourier transform and provide applications of its use

Dirichlet conditions for Fourier transforms

If a signal

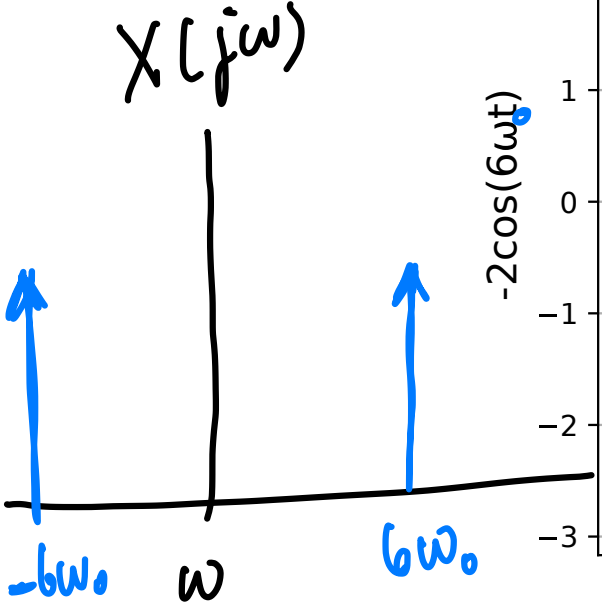
1. is single-valued
2. is absolutely integrable ($\int_{-\infty}^{\infty} |x(t)|dt < \infty$)
3. has a finite number of maxima and minima within any finite interval
4. has a finite number of finite discontinuities within any finite interval

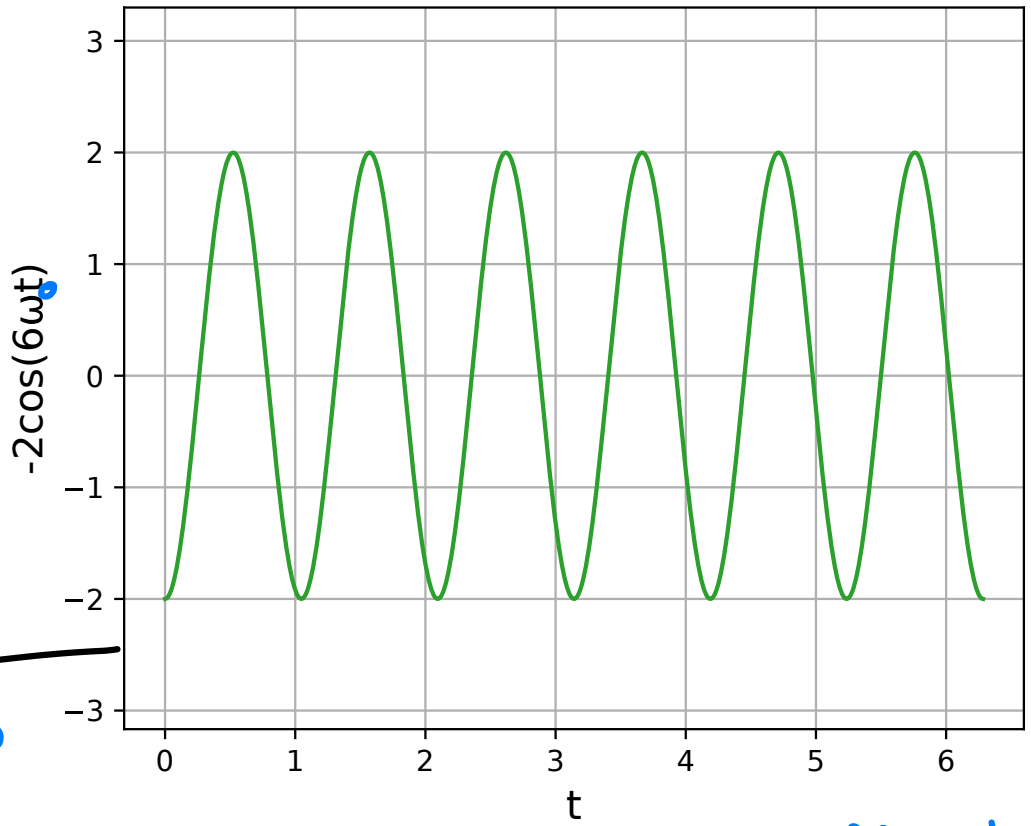
then the Fourier transform converges to

- $x(t)$ where it is continuous
- the average of the values on either side at a discontinuity

Dirichlet conditions for Fourier transforms

The "**within any finite interval**" takes care of periodic signals:



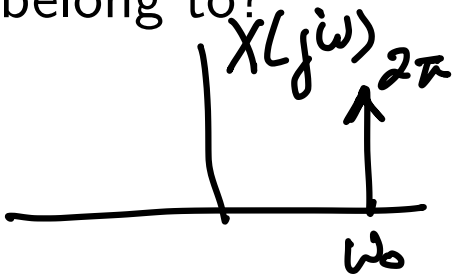


$$\cos(6\omega_0 t) = \frac{e^{j6\omega_0 t} + e^{-j6\omega_0 t}}{2}$$

Exercise 1

What signal does the following Fourier transform belong to?

$$X(j\omega) = 2\pi\delta(\omega - \omega_0)$$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\,\delta(\omega-\omega_0)\, e^{j\omega t}\, d\omega$$

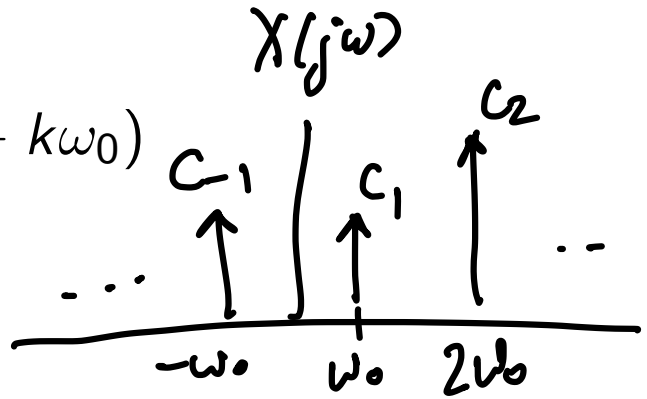
$$= \int_{-\infty}^{\infty} \delta(\omega-\omega_0)\, e^{j\omega t}\, d\omega$$

$$= e^{j\omega_0 t}$$

Exercise 2

What signal does the following Fourier transform belong to?

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)$$



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

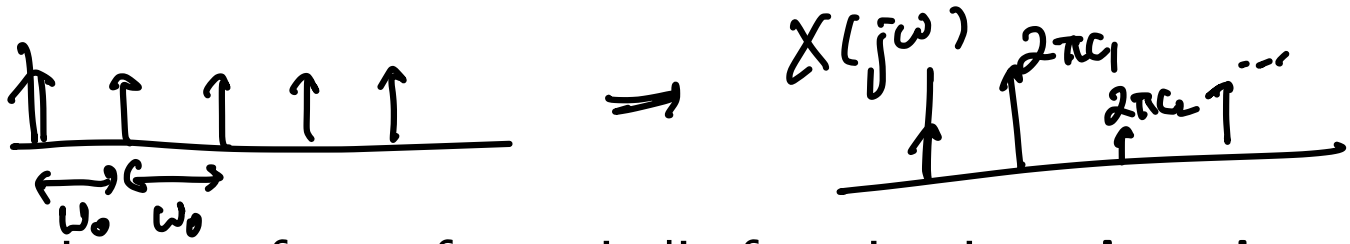
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)\, e^{j\omega t} d\omega$$

$$= \sum_{k=-\infty}^{\infty} c_k \int_{-\infty}^{\infty} \delta(\omega - k\omega_0)\, e^{j\omega t} d\omega$$

$$= \sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}$$

→ Fourier Synthesis!

Fourier transforms for periodic signals: a unified representation



The Fourier transform of a periodic function is an **impulse train**.

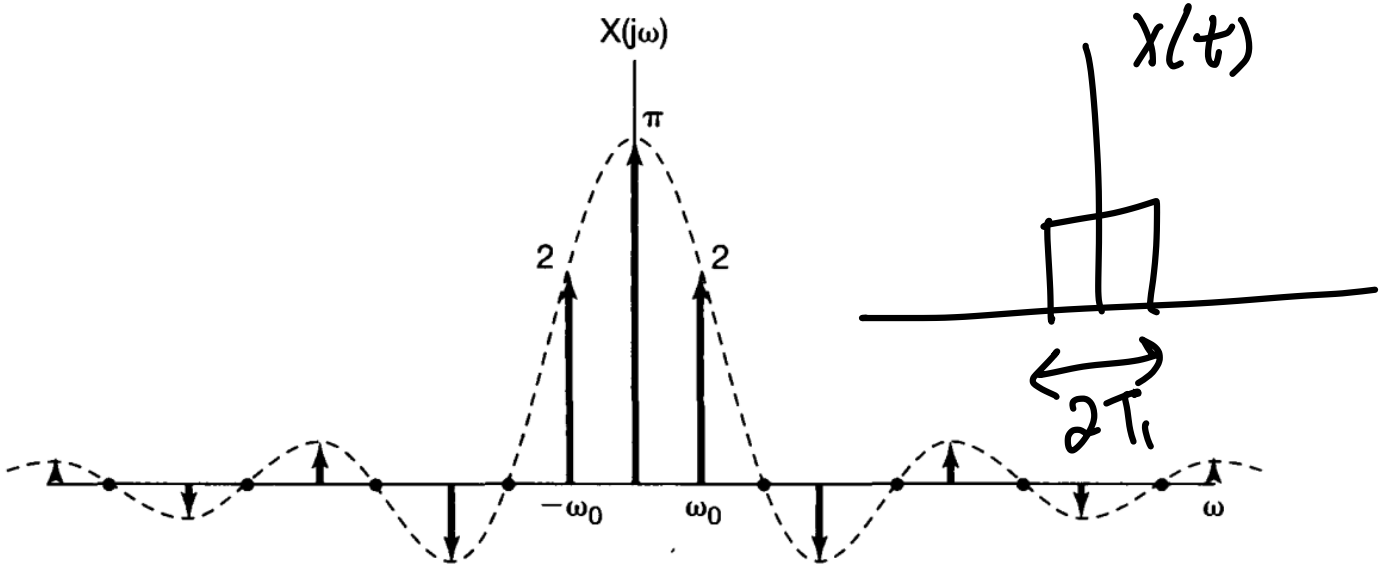
$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)$$

The impulses have area $2\pi c_k$ and are positioned at the harmonically related frequencies.

Fourier transforms for periodic signals: a unified representation

Its Fourier *transform* will be

$$X(j\omega) = 2\pi c_0 \delta(\omega) + \sum_{k=-\infty}^{\infty} 2\pi \frac{2\sin(k\omega_0 T_1)}{k\omega_0 T} \delta(\omega - k\omega_0)$$



(Plotted for $T = 4T_1$)

Image credit: Oppenheim chapter 4.2

Fourier transform properties

I'll provide some notes / exercises to practice applying these properties.

"Tutorial 5.5"

TABLE 4.1 PROPERTIES OF THE FOURIER TRANSFORM

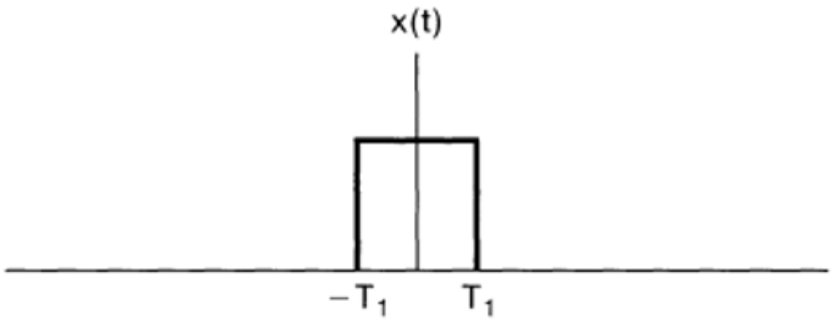
Section	Property	Aperiodic signal	Fourier transform
		$x(t)$ $y(t)$	$X(j\omega)$ $Y(j\omega)$
4.3.1	Linearity	$ax(t) + by(t)$	$aX(j\omega) + bY(j\omega)$
4.3.2	Time Shifting	$x(t - t_0)$	$e^{-j\omega t_0}X(j\omega)$
4.3.6	Frequency Shifting	$e^{j\omega_0 t}x(t)$	$X(j(\omega - \omega_0))$
4.3.3	Conjugation	$x^*(t)$	$X^*(-j\omega)$
4.3.5	Time Reversal	$x(-t)$	$X(-j\omega)$
4.3.5	Time and Frequency Scaling	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
4.4	Convolution	$x(t) * y(t)$	$X(j\omega)Y(j\omega)$
4.5	Multiplication	$x(t)y(t)$	$\frac{1}{2\pi}\int_{-\infty}^{+\infty}X(j\theta)Y(j(\omega - \theta))d\theta$
4.3.4	Differentiation in Time	$\frac{d}{dt}x(t)$	$j\omega X(j\omega)$
4.3.4	Integration	$\int_{-\infty}^{t}x(t)dt$	$\frac{1}{j\omega}X(j\omega) + \pi X(0)\delta(\omega)$
4.3.6	Differentiation in Frequency	$tx(t)$	$j\frac{d}{d\omega}X(j\omega)$
4.3.3	Conjugate Symmetry for Real Signals	$x(t)$ real	$\begin{cases} X(j\omega) = X^*(-j\omega) \\ \mathcal{R}e\{X(j\omega)\} = \mathcal{R}e\{X(-j\omega)\} \\ \mathcal{I}m\{X(j\omega)\} = -\mathcal{I}m\{X(-j\omega)\} \\ \lvert X(j\omega)\rvert = \lvert X(-j\omega)\rvert \\ \measuredangle X(j\omega) = -\measuredangle X(-j\omega) \end{cases}$
4.3.3	Symmetry for Real and Even Signals	$x(t)$ real and even	$X(j\omega)$ real and even
4.3.3	Symmetry for Real and Odd Signals	$x(t)$ real and odd	$X(j\omega)$ purely imaginary and odd
4.3.3	Even-Odd Decomposition for Real Signals	$x_e(t) = \mathcal{E}v\{x(t)\}$ [$x(t)$ real] $x_o(t) = \mathcal{O}d\{x(t)\}$ [$x(t)$ real]	$\mathcal{R}e\{X(j\omega)\}$ $j\mathcal{I}m\{X(j\omega)\}$

Image credit: Oppenheim chapter 4.6

Time/frequency duality of the FT

Let's consider a single square pulse:

$$x(t) = \begin{cases} 1 & |t| < T_1 \\ 0 & |t| > T_1 \end{cases}$$



Its Fourier spectrum is

$$X(j\omega) = 2\,\frac{\sin(\omega T_1)}{\omega}$$

Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

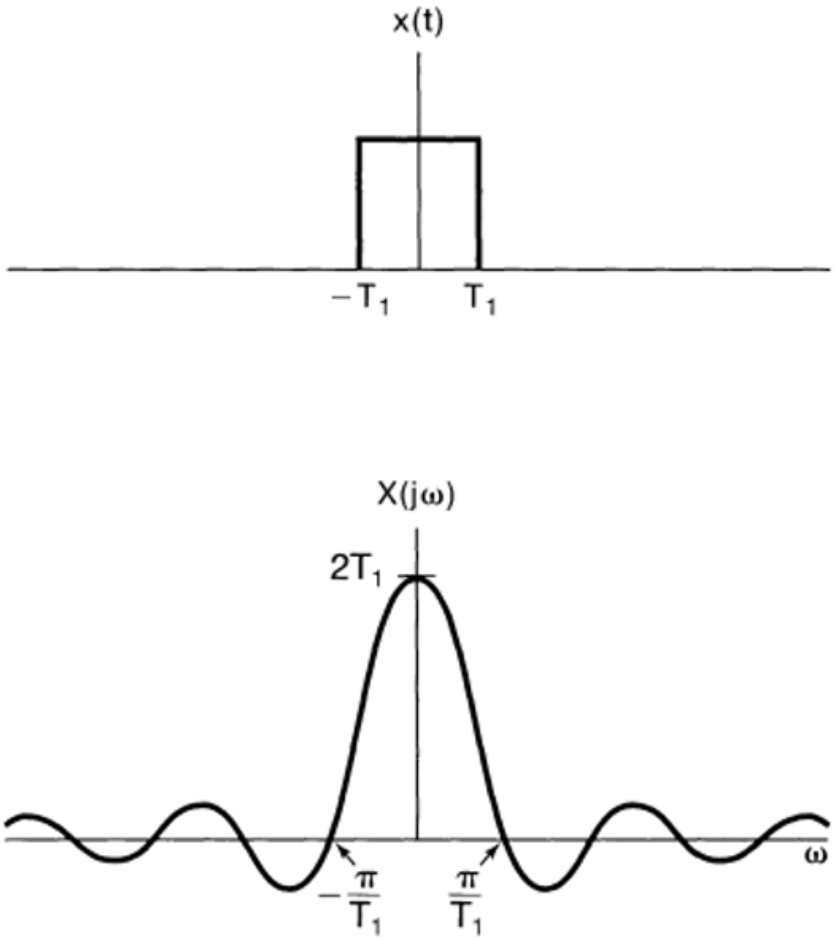


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Now let's consider a signal whose Fourier transform is

$$X(j\omega) = \begin{cases} 1 & |\omega| < W \\ 0 & |\omega| > W \end{cases}$$

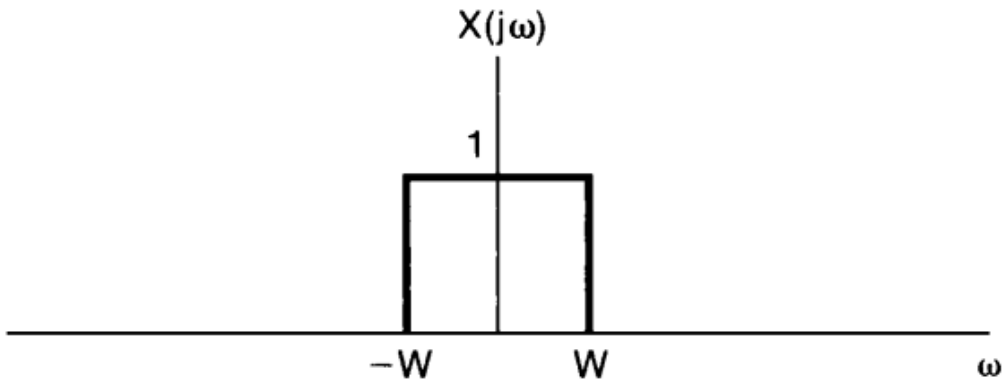


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Compute the inverse Fourier transform:

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

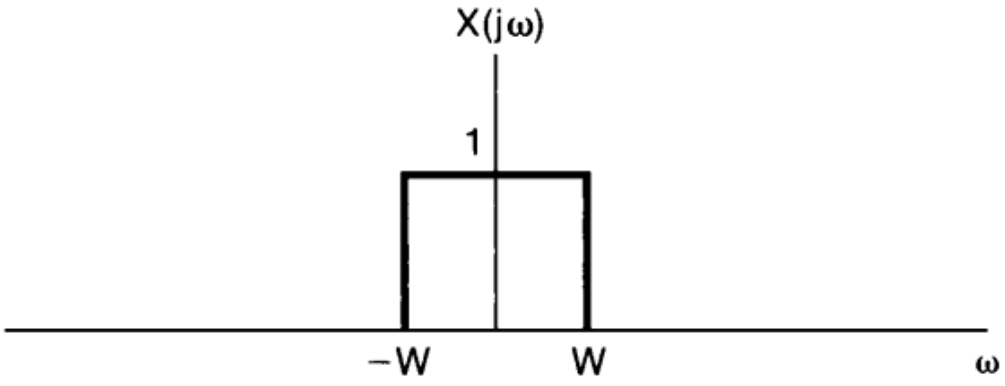
$$= \frac{1}{2\pi}\int_{-W}^{W} e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi jt}\, e^{j\omega t}\Big|_{-W}^{W}$$

$$= \frac{1}{2\pi jt}\left[e^{jWt} - e^{-jWt}\right]$$

$$= \frac{1}{2\pi jt}\cdot 2j\sin(Wt) = \frac{\sin(Wt)}{\pi t}$$

Time/frequency duality of the FT



$$H(j\omega) = \int_{-\infty}^{\infty} h(t)\, e^{-j\omega t}\, dt$$

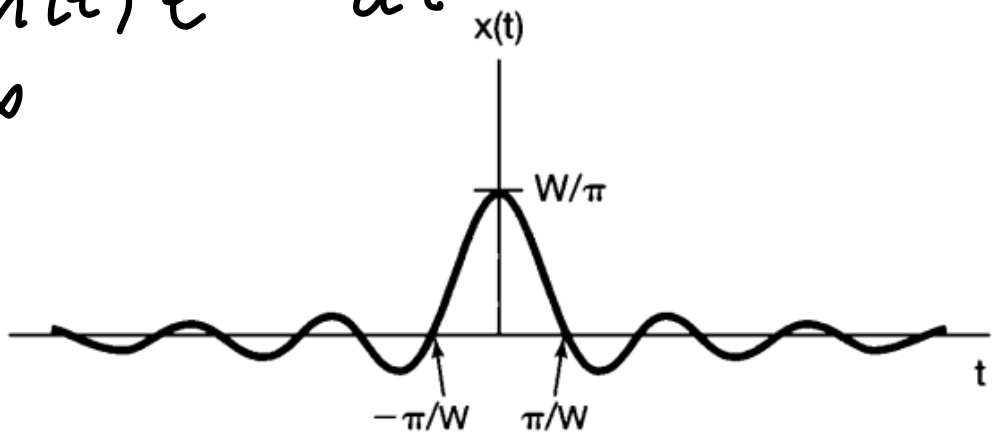


Image credit: Oppenheim chapter 4.1

Time/frequency duality of the FT

Duality: for any transform pair ($x(t) \leftrightarrow X(j\omega)$), there is a *dual pair* with the time and frequency variables interchanged.

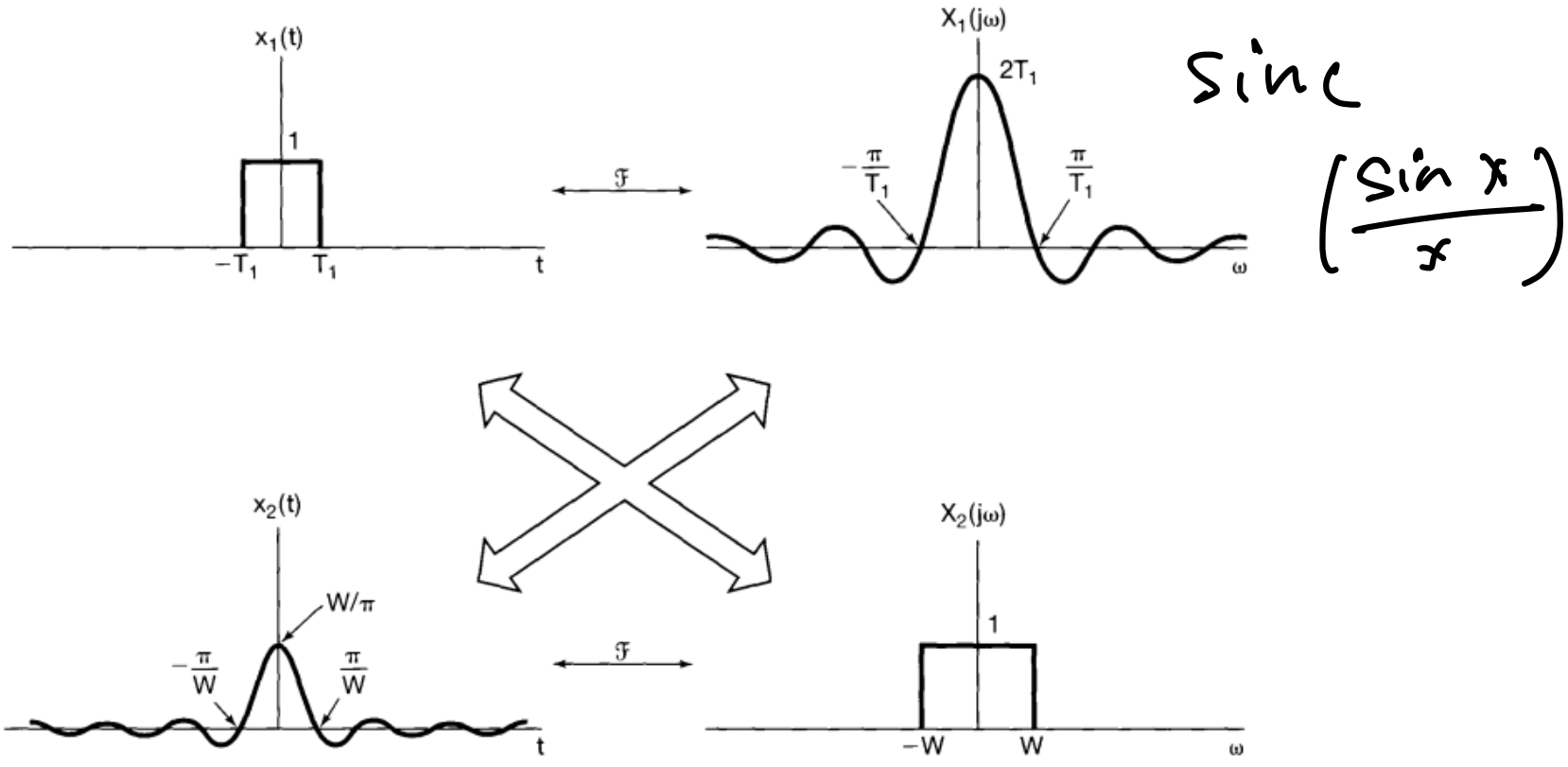


Image credit: Oppenheim chapter 4.3

Convolution and the Fourier transform

Recall complex exponentials are eigenfunctions of LTI systems. If we input signal $x(t)$ into LTI system with impulse response $h(t)$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$y(t) = \sum_{k=-\infty}^{\infty} c_k H(jk\omega) e^{jk\omega t}$$

where

$$H(j\omega) = \int_{-\infty}^{\infty} h(t) e^{-j\omega t} dt$$

This came from the convolution integral:

$$y(t) = \int_{-\infty}^{\infty} x(t-\tau) h(\tau) d\tau = x(t) \cdot H(j\omega)$$

Convolution and the Fourier transform

Let's express $x(t)$ using the inverse Fourier transform:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

and put this into the convolution integral...

$$\begin{aligned}
y(t) &= \int_{-\infty}^{\infty} x(t-\tau)\, h(\tau)\, d\tau \\
&= \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega(t-\tau)}\, d\omega \right] h(\tau)\, d\tau \\
&= \int_{-\infty}^{\infty} \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} \right] e^{-j\omega\tau} h(\tau)\, d\tau\, d\omega \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} \left[\int_{-\infty}^{\infty} e^{-j\omega\tau} h(\tau)\, d\tau \right] d\omega \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)\, H(j\omega)\, e^{j\omega t}\, d\omega
\end{aligned}$$

Convolution and the Fourier transform

We have **two** ways to write $y(t)$:

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\,H(j\omega)\,e^{j\omega t}\,d\omega \quad \text{(last slide)}$$

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(j\omega)\,e^{j\omega t}\,d\omega \quad \text{(inverse FT)}$$

This has an important implication:

$$Y(j\omega) = H(j\omega)\,X(j\omega)$$

$$y(t) = h(t) * x(t)$$

For next time

Content:

- Behaviour of the Fourier transform under differentiation and integration
- LTI systems based on differential equations

Action items:

1. Tutorial Assignment 3 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 4.4-4.6
- Suggested problems: 4.4, 4.6, 4.9, 4.12, 4.15, 4.17, 4.19, 4.26, 4.32
- For next class: Oppenheim chapter 4.7, 6.1-6.2